

PERSONAL INFORMATION

Andrea Silverj, Ph.D.



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X @silverjand

CURRENT POSITION

Postdoctoral Research Fellow at the University of Calgary

ABOUT ME

I'm a computational evolutionary biologist specialised in viral metagenomics and phylogenomics. I study phages of the human microbiome and arboviruses, such as West Nile virus. Interested in mobile genetic elements, such as plasmids and transposons. I'm currently developing AI-based pipelines for virus discovery and evolutionary analysis.

WORKING EXPERIENCES

Jan 2025 – Ongoing

Postdoctoral Research Fellow

University of Calgary, Canada

Project:

- "Structure-guided discovery and evolutionary analysis of viruses"

Funding: VPR Postdoctoral Match-Funding and Beverley Phillips Postdoctoral Fellowships

Supervisor: Prof. Ann C. Gregory

Aug 2025

Visiting Research Scientist

Kasitsna Bay Laboratory, Alaska, United States

National Centers for Coastal Ocean Science – NOAA

Co-supervisor for the Project:

- "Investigating transport feasibility, ecological, and public health implication of glacial melt microbes in Alaska"

Funding: University of Calgary

Supervised student: Harland Ammentorp

Supervisor: Prof. Ann C. Gregory

Dec 2023 – Nov 2024

Research Fellow

Alma Mater Studiorum – University of Bologna, Italy

Projects:

- "Phylogenomics and comparative genomics in mosquitoes"

Funding: NextGenerationEU/PNRR fellowship

- "Genome plasticity in tiger mosquitoes: biological significance and relevance for pest management"

Funding: Prin 2022 fellowship

Supervisor: Prof. Andrea Luchetti

Mar 2018 – Mar 2019

Internship

Edmund Mach Foundation (Italy) – Agrarian Entomology laboratory

Supervisor: Prof. Omar Rota-Stabelli

- Molecular evolution
- Arboviruses
- R and Python programming

EDUCATION AND TRAINING

Nov 2019 – Mar 2024

Ph.D. in Agrifood and Environmental Sciences

University of Trento, Italy

Thesis title: "Viral metagenomics and phylogenomics for One Health"

Supervisors: Prof. Omar Rota-Stabelli and Prof. Nicola Segata

Examination committee: Prof. Mang Shi (Sun Yat-Sen University, China), Dr. Laura Treu (University of Padova, Italy), Dr. Cristina Menni (King's College London, United Kingdom), Prof. Gerardo Puopolo (University of Trento, Italy)

Final grade: *Summa cum laude*

October 2016 – March 2019

Master of Science (M.Sc.) in Evolutionary Biology

University of Padova, Italy

Thesis title: "An evolutionary analysis of codon usage bias in Zika virus to identify possible vectors and host"

Supervisor: Prof. Giuseppe Fusco

Final grade: 110L – *Summa cum laude*

October 2012 – March 2016

Bachelor of Science (B.Sc.) in Biological Sciences

University of L'Aquila, Italy

Thesis title: "Aging in vertebrates: an evolutionary point of view"

Supervisor: Prof. Adriano Angelucci

Final grade: 110L – *Summa cum laude*

PERSONAL SKILLS

Mother tongue(s)

Italian

Other language(s)

	UNDERSTANDING		SPEAKING		WRITING
	Listening	Reading	Spoken interaction	Spoken production	
English	C2	C2	C2	C2	C2

Ph.D. course held in English, involving numerous participations at conferences, and writing of scientific papers.

Levels: A1/A2: Basic user - B1/B2: Independent user - C1/C2: Proficient user

[Common European Framework of Reference for Languages](#)

Position-related skills

- Experience with large-scale analyses of biological sequence data, molecular evolution, comparative methods, metagenomics, phylogenomics and molecular clocks using maximum likelihood and Bayesian methods.

Digital skills

- UNIX shell scripting
- R and Python programming
- AI and machine learning
- Git
- HPC computing; Containers (Singularity/Apptainer, Docker)
- HTML and CSS
- GIS
- Advanced command of Microsoft Office™ tools
- Very good knowledge of Windows/Linux/macOS operative systems
- Inkscape, Adobe Illustrator, Adobe Photoshop
- Most used softwares: BEAST, PhyloBayes, IQ-TREE, RAxML, MAFFT, bioBakery tools, vConTACT2, VirSorter2, CheckV, OrthoFinder, RDP4, Prokka, BLAST, Bowtie2, BRAKER, SPAdes, MaSuRCA, BUSCO, BlobTools, AlphaFold, Foldseek

- Developed software
- SiDcal (<https://github.com/andrea-silverj/SiDcal>)
 - MetaPhlAn v4.1 – Viral module (<http://segatalab.cibio.unitn.it/tools/metaphlan/index.html>)
- Scientific interests
- Evolutionary biology, phylogenomics, viral metagenomics, computational biology, bioinformatics, AI, mobilome, philosophy of biology

ADDITIONAL INFORMATION

Memberships

- Member of the Canadian Society for Microbiologists (since 2026)
- Member of the International Society for Microbial Ecology (since 2026)
- Pacific Organizing Committee Member – Microbiome Virtual International Forum (since 2025)
- Member of the American Society for Microbiology (since 2025)
- Member of the Society for Molecular Biology and Evolution (since 2019)
- Member of the Italian Federation of Life Sciences (since 2019)
- Member of the Italian Society for Evolutionary Biology (since 2017)

Awards

- Peoples' Choice Poster in Poster Session 2 at the Having IMPACTT 5: Advancing Microbiome Research Symposium – Canmore, Alberta, Canada (200 CAD)
- Best talk at the XVI FISV congress (2022) – Environmental, Evolutionary and Developmental Biology section
- Winner of the “Golden Charley Prize” for the best creative performance in the outreach of Evolutionary Biology at the 8th congress of the Italian Society for Evolutionary Biology (2019)

Outreach

- Collaborative session on the iteration of science centre galleries – Telus Spark Science Centre in Calgary, Canada (2025)
- European Researcher's Night at the MUSE Science Museum in Trento, Italy
 - Molecular clocks to study the evolution of life on Earth (2021)
 - Outreaching the behaviour of *Drosophila suzukii* (2018)

Organised workshops

- Co-organiser of the Bioinformatics Crash Course 2026 at the University of Calgary, coordinating the activities of more than 50 participants. All course practicals were conducted on the TALC HPC infrastructure of the University of Calgary
- MVIF #46 – Organizer and keynote speaker of the event, introducing and moderating Dr. Simon Roux
- MVIF #44 – Organizer and keynote speaker of the event, introducing and moderating Prof. Katherine P. Lemon
- Co-founder and organiser of the Bioinformatics Working Group at the University of Calgary, coordinating more than 90 people with different backgrounds and level of experience in coding
- [SIBE2024 Crash Course](#) – Inferring phylogenies from metagenomic data
- Founder and organiser of the [Italian Phylogenetics Workshop \(ITA*PHY\)](#)
The first edition, held in June 2023, has seen more than 30 participants from different parts of the world, with three invited speakers (Sandra Álvarez Carretero, Benoît Morel and Omar Rota-Stabelli)

Driving license

- Italian driving licence (B)

References

- Prof. Ann C. Gregory, Ph.D. E-mail: ann.gregory@ucalgary.ca
- Prof. Omar Rota-Stabelli, Ph.D. E-mail: omar.rota@fmach.it
- Prof. Nicola Segata, Ph.D. E-mail: nicola.segata@unitn.it
- Prof. Andrea Luchetti, Ph.D. E-mail: andrea.luchetti@unibo.it
- Annapaola Rizzoli, Ph.D. E-mail: annapaola.rizzoli@fmach.it
- Prof. Giuseppe Fusco, Ph.D. E-mail: giuseppe.fusco@unipd.it

List of publications

Published in peer-reviewed journals

- [1] Silverj* A., Rota-Stabelli O. (2020) On the correct interpretation of similarity index in codon usage studies: comparison with four other metrics and implications for Zika and West Nile virus. *Virus Research*. 286, 198097. <https://doi.org/10.1016/j.virusres.2020.198097>
- [2] Bianco L., Moser M., Silverj A., Micheletti D., Lorenzin G., Collini L., Barbareschi M., Lanzafame P., Segata N., Pindo M., Franceschi P., Rota-Stabelli O., Rizzoli A., Fontana P., Donati C. (2022) On the origin and propagation of the COVID-19 outbreak in the Italian province of Trento, a tourist region of Northern Italy. *Viruses*. Mar 11;14(3):580. <https://doi.org/10.3390/v14030580>
- [3] Mencattelli* G., Silverj* A., Iapaolo F., Ippoliti C., Teodori C., Di Gennaro A., Curini V., Candeloro L., Conte A., Polci A., Morelli D., Perrotta M.G., Marini G., Rosà R., Monaco F., Segata N., Rizzoli A., Savini G., West Nile Working Group (2022) Epidemiological and evolutionary analysis of West Nile virus lineage 2 in Italy. *Viruses*. 15(1), 35. <https://doi.org/10.3390/v15010035>
- [4] Marchesini A., Silverj A., Torre S., Rota-Stabelli O., Girardi M., Passeri I., Fracasso I., Sebastiani F., Vernesi C. (2023) First genome-wide data from Italian European beech (*Fagus sylvatica* L.): Strong and ancient differentiation between Alps and Apennines. *PLOS ONE*. 18(7): e0288986 <https://doi.org/10.1371/journal.pone.0288986>
- [5] Mencattelli* G., Ndione* H.D., Silverj* A., Diagne M.M., Curini V., Teodori L., Di Domenico M., Mbaye R., Leone A., Marcacci M., Gaye A., Ndiaye E., Diallo D., Ancora M., Secondini B., Di Lollo V., Mangone I., Bucciacchio A., Polci A., Marini G., Rosà R., Segata N., Fall G., Cammà C., Monaco F., Diallo M., Rota-Stabelli O., Faye O., Rizzoli A., Savini G. (2023) Spatial and temporal dynamics of West Nile virus between Africa and Europe. *Nature communications*. 14, 6440. <https://doi.org/10.1038/s41467-023-42185-7>
- [6] Zadra N., Tatti A., Silverj A., Piccinno R., Devilliers J., Lewis C., Arnoldi D., Montarsi F., Escuer P., Fusco G., De Sanctis V., Feuda R., Sánchez-Gracia A., Rizzoli A. and Rota-Stabelli O. (2023) Genome skimming of *Aedes japonicus* and *Aedes koreicus* from Italy and an updated picture of their evolution based on mitogenomics and barcoding. *Insects*. 14(12), 904. <https://doi.org/10.3390/insects14120904>
- [7] Silverj* A., Mencattelli* G., Monaco F., Iapaolo F., Teodori L., Leone A., Polci A., Curini V., Di Domenico M., Secondini B., Di Lollo V., Ancora M., Di Gennaro A., Morelli D., Perrotta M.G., Marini G., Rosà R., Segata N., Rota-Stabelli O., Rizzoli A., Savini G. (2024) Origin and evolution of West Nile virus lineage 1 in Italy. *Epidemiology&Infection*. 152, e150, 1-11. <https://doi.org/10.1017/S0950268824001420>
- [8] Verni S., Langeneck J., Silverj A., Stefanni S., Musco L., Castelli A. (2025) Diversity and biogeography of mesophotic and bathyal Onuphidae (Annelida: Eunicida) of the Western Mediterranean Sea. *Deep-Sea Research Part I*. <https://doi.org/10.1016/j.dsr.2025.104476>

Submitted, accepted or preprint available

- [9] Pinto F., Zolfo M., Beghini F., Armanini F., Asnicar F., Silverj A., Boscaini A., Salmaso N., Segata N. (2020) A step-by-step sequence-based analysis of virome enrichment protocol for freshwater and sediment samples. Applied and Environmental Microbiology. *Preprint available on bioRxiv*: <https://doi.org/10.1101/2020.09.17.302836>
- [10] Zolfo M., Silverj A., Blanco-Míguez A., Manghi P., Rota-Stabelli O., Heidrich V., Jensen J., Maharjan S., Franzosa E., Menni C., Visconti A., Pinto F., Ciciani M., Huttenhower C., Cereseto A., Asnicar F., Kitano H., Yamada T., Segata N. (2024) Discovering and exploring the hidden diversity of human gut viruses using highly enriched virome samples. *Preprint available on bioRxiv*: <https://doi.org/10.1101/2024.02.19.580813>. *Submitted in Nature Biotechnology*.
- [11] Ndiaye M., Balde D., Efire A., Diouf L., Dia N., Cisse M., Cunha M., Silverj A., Mboowa G., Fall G., Dia N., Faye O., Amadou Sall A., Berthet X., Moussa Diagne M., Weidmann M., Faye O., Dieng I. (2024) Global In-Silico Performance Assessment of the CDC Zika Triplex Detection System. *Submitted in Microbiology spectrum*.
- [12] Silverj* A., Alfano N., Tagliapietra V., Rosso F., Segata N., Rota-Stabelli O. and Rizzoli A. Evolutionary analysis of viral metagenome-assembled genomes from Italian tick populations detects pathogens and reveals new viral species. *Submitted in Discover Viruses*.

In preparation

- [13] Silverj* A., Zolfo M., Asnicar F., Blanco-Míguez A., Cumbo F., Huang K.D., Pinto F., Rota-Stabelli O., Segata N. Large-scale genome reconstructions from human gut metagenomes to study phage-host relationships.
- [14] Silverj* A., Zadra N., Martellosi J., Tatti A., Battista M., Fouani J., Scala M., Roselli G., Pedrazzoli F., Vicelli B., Pindo M., Anfora G., Mazzoni V., Luchetti A. and Rota-Stabelli O. The short-read genome of two populations of the parasitoid wasp *Trissolcus japonicus* from Italy: male haploidy, genome of harbored *Wolbachia*, and identical mitogenomes.
- [15] Silverj* A., de Andrade A.A.S., Ost K., Gregory A.C. Discovery of putative active and cryptic viral infections in prevalent gut fungi.

Note: the symbol “*” indicates first or co-first authorship.

Conferences

IMPACTT 5 Symposium: Advancing Microbiome Research

22/09/2025 – 24/09/2025 – Canmore (Canada)

“Discovery of putative active and cryptic viral infections in prevalent gut fungi.”

Poster presentation.

10th SIBE/ISEB (Italian Society for Evolutionary Biology) congress

08/09/2024 – 11/09/2024 – Naples (Italy)

"Evolution of West Nile virus between Europe and Africa."

Oral presentation.

9th International Conference on Emerging Zoonoses

09/06/2024 – 12/06/2024 – Palermo (Italy)

"Metatranscriptomics of Italian tick populations reveals local virome biodiversity and new viral species."

Oral presentation.

SMBE2023

19/03/2023 – 21/03/2023 – Ferrara (Italy)

"Evolutionary analysis of viral metagenome-assembled genomes from Italian tick populations sampled on a latitudinal gradient."

Poster presentation.

ARES/EVA-G joint session

19/03/2023 – 21/03/2023 – Padua (Italy)

"Evolutionary analysis of viral metagenome-assembled genomes from Italian tick populations." and "Eco-epidemiological and genetic evolutionary features of WNV L1 and L2 in Italy."

Oral presentation as invited speaker.

Cell Symposium: Viruses in health and disease

19/03/2023 – 21/03/2023 – Sitges (Spain)

"Large-scale genome reconstructions from human gut metagenomes to study phage-host relationships."

Oral presentation.

9th SIBE/ISEB (Italian Society for Evolutionary Biology) congress

04/09/2022 – 07/09/2022 – Ancona (Italy)

"Large scale genome reconstructions from human gut metagenomes reveal weak coevolution between phages and their bacterial hosts."

Oral presentation.

FISV2022 congress

09/2022 – Naples (Italy)

"Large scale genome reconstructions from human gut metagenomes reveal weak coevolution between phages and their bacterial hosts."

Oral presentation.

SMBEv2021

2021 – Online

"Large scale genome reconstructions from human gut metagenomes reveal weak coevolution between phages and their bacterial hosts."

Oral presentation.

Virtual FISV symposium on SARS-CoV-2 biology and COVID-19: current research and perspectives.

16 September 2020.

"Tracing the SARS-CoV-2 outbreak in Trentino using Nanopore sequencing".

Abstract.

8th SIBE/ISEB (Italian Society for Evolutionary Biology) congress

09/2019 – Padua (Italy)

"Codon usage indicates that amphibians, reptiles and birds are major hosts for Zika and other arboviruses: implications for epidemiology and surveillance."

Oral presentation.

EuroCitizen WG5 Meeting

04/2019 – Paris (France)

Project contribution.

XI European Congress of Entomology

07/2018 – Naples (Italy)

"Using codon usage bias to investigate the role of alternative vectors in the spread of Zika."

Poster presentation.

7th SIBE/ISEB (Italian Society for Evolutionary Biology) congress

28/08/2017 – 31/08/2017 – Rome (Italy)

"404 Error: (Evolutionary) Page Not Found."

Poster presentation.

Seminars, courses, workshops, summer schools

ViralZone and the Revolution in 3D Virus Structure

Organised by the Swiss Institute of Bioinformatics (SIB) and held online on 27/03/2025

Emerging viral threats in a globalized society: molecular, epidemiological, clinical and social aspects of emerging viral diseases

Hosted by the University of Pavia and held online from 06/09/2021 to 10/09/2021

High-Performance Computing summer School 2021

Held in Trento, Italy, from August 30th to September 3rd, 2021 (24 hours)

DrosEU workshop

Held online on 21/06/2021 and 22/06/2021

I Simposio Virtuale FISV "Life Science and Society"

Held online on 28/04/2021

Molecular Phylogenetics: from genes to communities to kingdoms

Hosted by the University of Pavia and held online from 26/04/2021 to 30/04/2021

AGS Virtual Symposium Spring 2021

Held online on 13/04/2021

Large-scale Biological Network Analysis and Visualization Using CyVerse

Held online on 05/03/2021

Hit the Ground Running with SNP Calling for PopGen and Evolutionary Analyses

Held online on 11/09/2020

Laboratory health and safety (LHS) - Specific training

Held at Edmund Mach Foundation by C3A and FEM health and safety officers and consultants in 01/2020 (8 hours)

Bridging Phylogeny and Population genetics: Inferring divergence and selection at both interspecies and intraspecies level (summer school)

Held at the Campus of Fondazione Edmund Mach in San Michele all'Adige from 27/08/2019 to 01/09/2019

Fellowships and funding

[June – July 2025] Travel Grant to attend the 2025 HIDA CIFAR Summer School on Data Science & Climate (2,300 CAD)

[2025 – 2028] Beverley Phillips Postdoctoral Fellowship, Snyder Institute for Chronic Diseases (30,000 CAD for three years, for a total of 90,000 CAD)

[2025 – 2027] VPR Postdoctoral Match-Funding Fellowship, University of Calgary (25,000 CAD for two years, for a total of 50,000 CAD)

[2023 – 2024] NextGenerationEU/PNRR Research fellowship – Phylogenomics and comparative genomics in mosquitoes. (22, 699, 02 EUR)

[2023] SIBE workshop funds (2, 000 EUR)

[2019 – 2023] PhD fellowship – Edmund Mach Foundation (52, 942 EUR)

Teaching and supervision

[May 2025 – Ongoing] Bachelor's internship co-supervisor of a project on the diversity and evolution of DNA binding proteins in viruses.

Student: Shihong Li (Mitacs program)

[February 2025 – Ongoing] Master's thesis co-supervisor of a project on the virome of Alaskan glaciers.

Student: Harland Ammentorp

[February 2025] Microbiology seminar at the faculty of Biological Sciences at the University of Calgary: "Viral metagenomics and phylogenomics for One Health."

[January 2025 – Ongoing] Master's thesis co-supervisor of a project on the microbiome and virome of vultures.

Student: Zakie Hage

[September 2024] Instructor at the SIBE2024 Crash Course

[April – May 2024] Teaching assistant at the University of Bologna

I taught students in the Master of Science programs "Biodiversity and Evolution" and "Bioinformatics" for the course "Molecular Phylogenetics" (6 CFU). Specifically, I led modules on Bayesian phylogenetics, including both practical and theoretical sessions (8 hours).

[June 2023] Instructor at the Italian Phylogenetics Workshop 2023

[2020 – 2021] Master's thesis co-supervision

Student: Alessia Tatti

Program: Evolutionary Biology

University: University of Padova, Italy

Thesis title: Phylogenetic approaches to study fungal biodiversity

[April 2021] Teaching assistant at the University of Trento

I taught undergraduate students in the "Enology and Viticulture" bachelor's program for the course "General Biology and Ecology" (12 CFU).

I was responsible for the "Virology" module (4 hours).